Propeller Bench Test Analysis Report

MN_4014_15x5

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Step 1 — Polynomial fits (throttle → RPM / Thrust / Torque)

Channel	Degree	R^2
rpm	1	0.99661
thrust	2	0.99935
torque	3	0.99965

Equations:

```
rpm(x) = 70.2593·x + 781.49
thrust(x) = 0.0020455·x^2 + 0.0960918·x - 0.361797
torque(x) =  - 2.94144e-07·x^3 + 7.56245e-05·x^2 + 0.00046387·x + 0.0104334
```

Step 2 — Lag lookup tables (see lag_vs_throttle_rate and lag_vs_rpm CSVs)

Step 3 — Lambda model: RPM_rate = λ·(RPM_cmd − RPM) | variant comparison

Variant	N	R2	Rmse	Nrmse	Λ Mean	Λ Std	Λ Min	Λ Max	Frac Negative
direct_all	9510	0.2415	3.7923	1.023	3.708	4.354	-8.871	20.721	0.225
direct_filtered	8951	0.3120	3.6370	1.033	3.522	4.385	-9.116	20.919	0.239
lag_all	11254	0.1430	3.7348	0.355	10.519	4.034	6.921	19.137	0.000
lag_filtered	10689	0.1704	3.7276	0.355	10.501	4.093	6.921	19.137	0.000

Methodology & Output Guide

Step 3 — Lambda Estimation Approaches

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The first-order RPM model is:  RPM_rate = λ · (RPM_commanded − RPM)
λ (lambda, s⁻¹) is the bandwidth coefficient. Two independent methods estimate it:

  DIRECT   Rearranges the model equation at every time step:
             λ = RPM_rate / (RPM_cmd − RPM)
           RPM_rate  = Savitzky-Golay numerical derivative of measured RPM.
           RPM_cmd   = global polynomial fit (Step 1) evaluated at current throttle.
           Requires |RPM_cmd − RPM| ≥ 50 RPM to avoid noisy near-steady-state estimates.
           Reflects the instantaneous, model-equation-based lambda at each sample.

  LAG      Inverts the empirical time lag between throttle input and RPM response:
             λ = 1 / τ  =  1000 / lag_rpm_ms
           lag_rpm_ms is measured in Step 2 via threshold-crossing analysis —
           the time for RPM to cross 30/40/50/60/70 % of its step range after
           the corresponding throttle crossing.  Treating this lag as the first-
           order time constant τ gives λ = 1/τ.  This is a system-identification
           estimate derived entirely from the measured response timing, independent
           of the model equation.
```

Lambda Fit Metrics Interpretation

```
  R²          Explained variance of the bivariate polynomial λ(RPM, RPM_rate).
              Higher is better.  < 0.3 means λ has little dependence on RPM / rate.
  RMSE        Root-mean-square prediction error (s⁻¹).  Lower is better.
  NRMSE       RMSE / |mean λ|.  Scale-free; < 0.5 indicates a well-conditioned fit.
  λ mean      Physically expect ~5–15 s⁻¹  (= 1/lag; lag ≈ 70–200 ms).
  λ std       Spread of observed λ values across the flight envelope.
  λ min/max   Check for physical plausibility — all values should be positive.
  neg %       Fraction of λ < 0 (unphysical); 0 % is ideal.
  filter Δ    Rows dropped by the |RPM_rate| threshold.  0 means threshold too high.
  |rate|max   Largest |RPM_rate| retained; compare to threshold to assess filter.
```

Output Files

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  global_fit_coeffs.csv          Polynomial coefficients: throttle → RPM / Thrust / Torque
  global_fit_results.txt         Human-readable fit report with equations and R²
  lag_vs_throttle_rate.csv       Weighted mean lag (ms) binned by throttle % × throttle rate %/s
  lag_vs_rpm.csv                 Thrust & torque lag (ms) binned by mean RPM
  lambda_lookup_<variant>.csv    Binned λ table (RPM × RPM_rate) for each variant
  lambda_fit_coeffs_<variant>.csv  Bivariate polynomial coefficients for λ(RPM, RPM_rate)
  lambda_comparison.csv          Side-by-side metrics for all four variants
  <motor>_report.pdf             This report (all figures + title + methodology pages)

  Variants:  direct_all / direct_filtered — direct approach, unfiltered / rate-limited
             lag_all    / lag_filtered    — lag-based approach, unfiltered / rate-limited
```
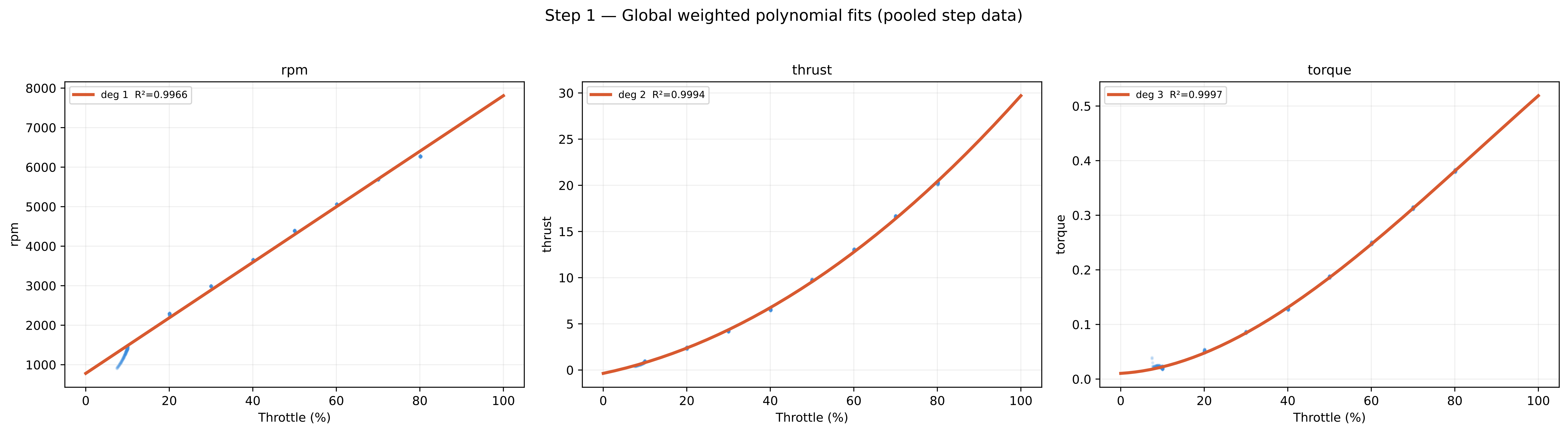
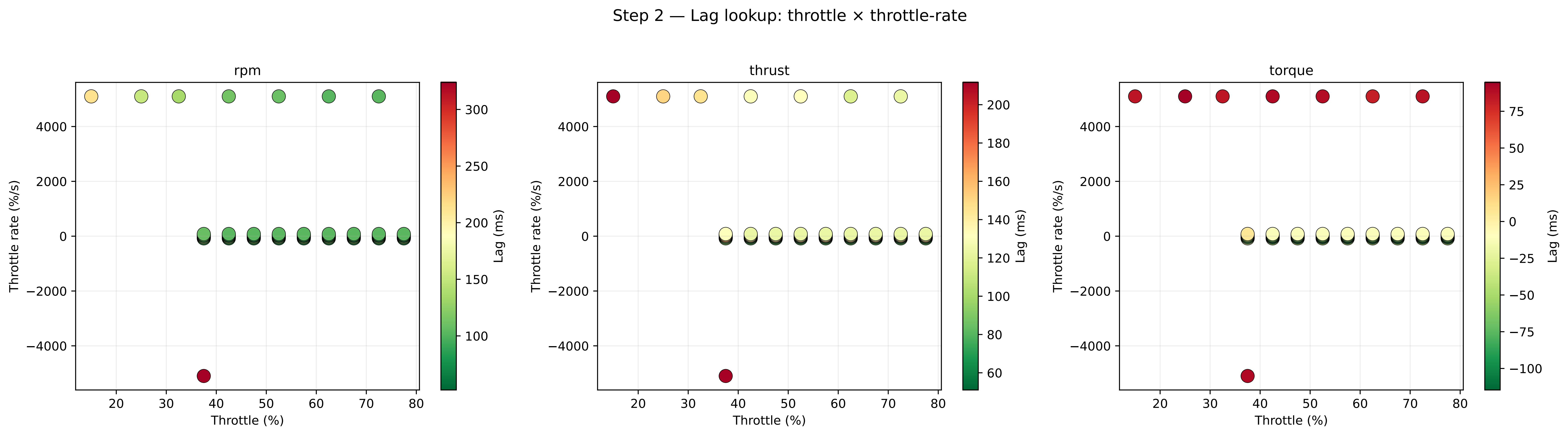
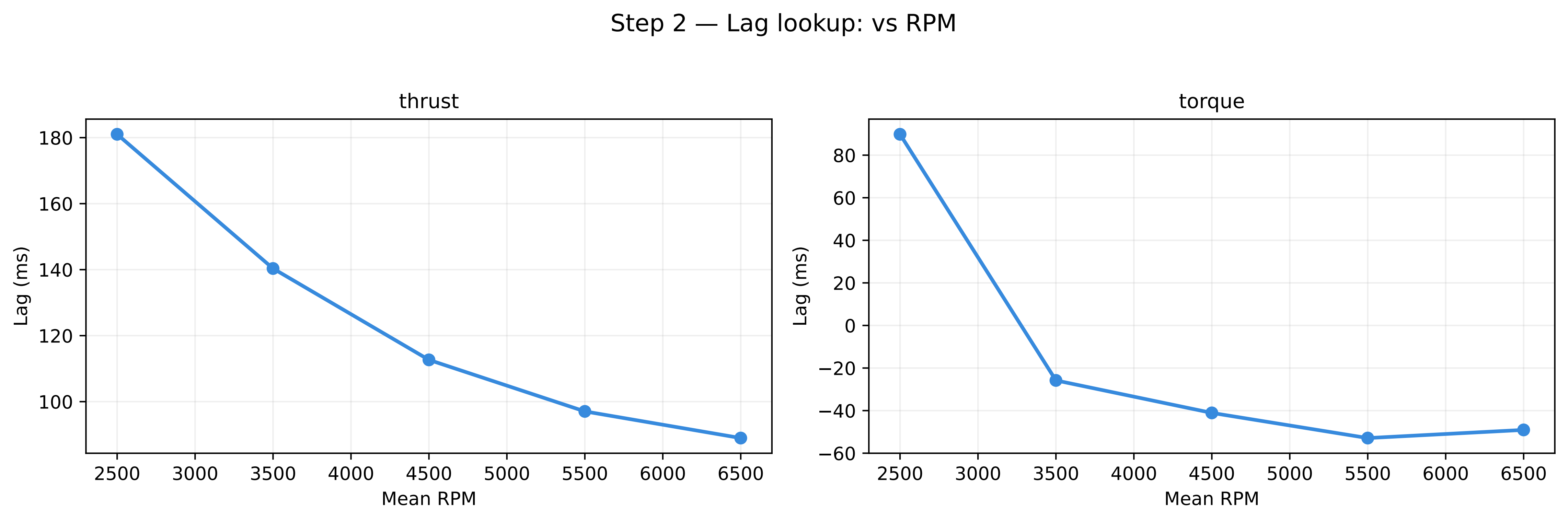
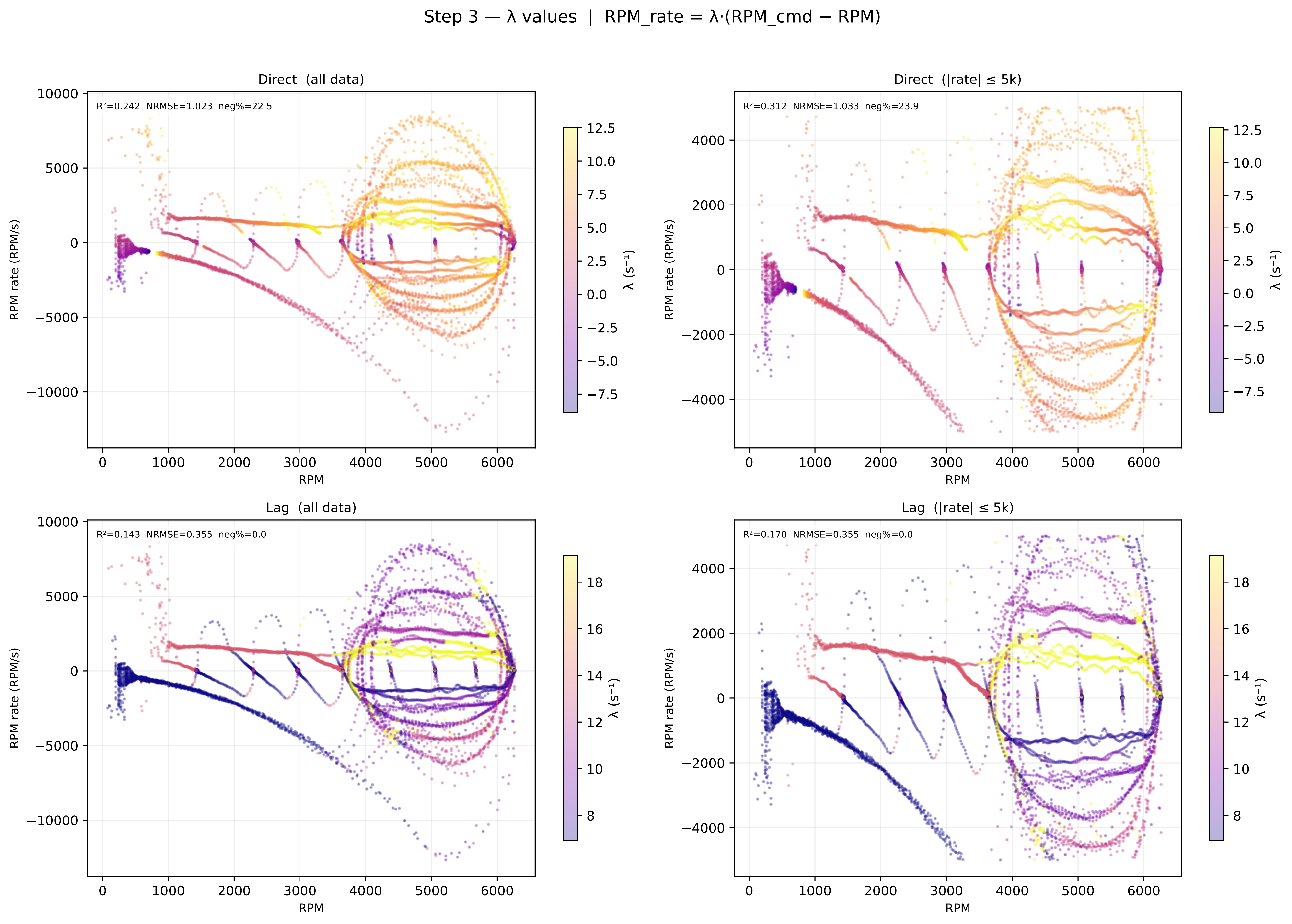
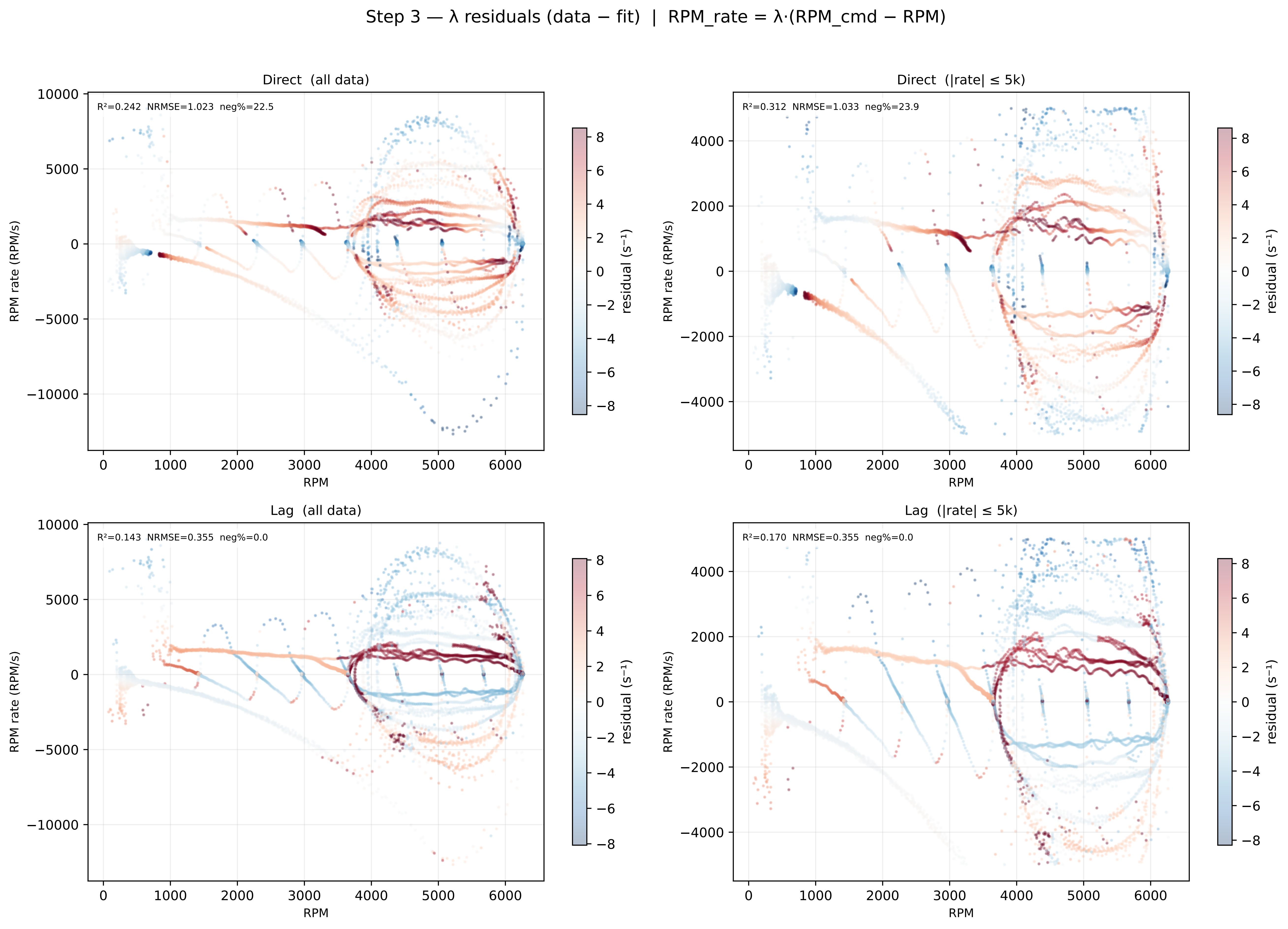



Step 1 — Global weighted polynomial fits (pooled step data)
rpm
deg 1 R²=0.9966
rpm
Throttle (%)
thrust
deg 2 R²=0.9994
thrust
Throttle (%)
torque
deg 3 R²=0.9997
torque
Throttle (%)

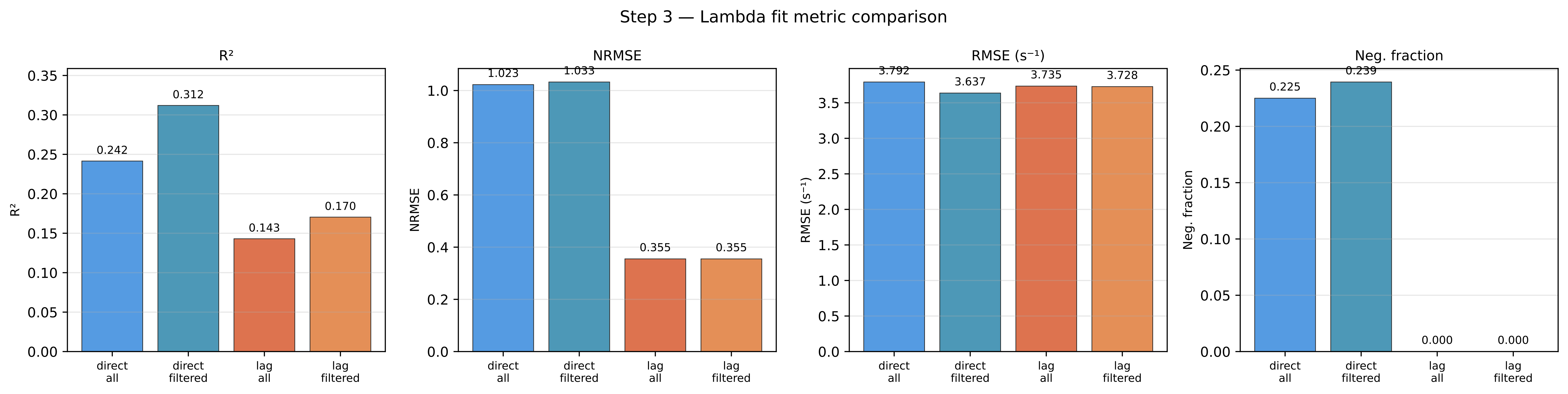

Step 2 — Lag lookup: throttle × throttle-rate
rpm
thrust
torque
Throttle rate (%/s)
Throttle (%)
Lag (ms)
4000
2000
0
−2000
−4000
20
30
40
50
60
70
80
300
250
200
150
100
180
160
140
120
80
60
75
50
25
−25
−50
−75
−100

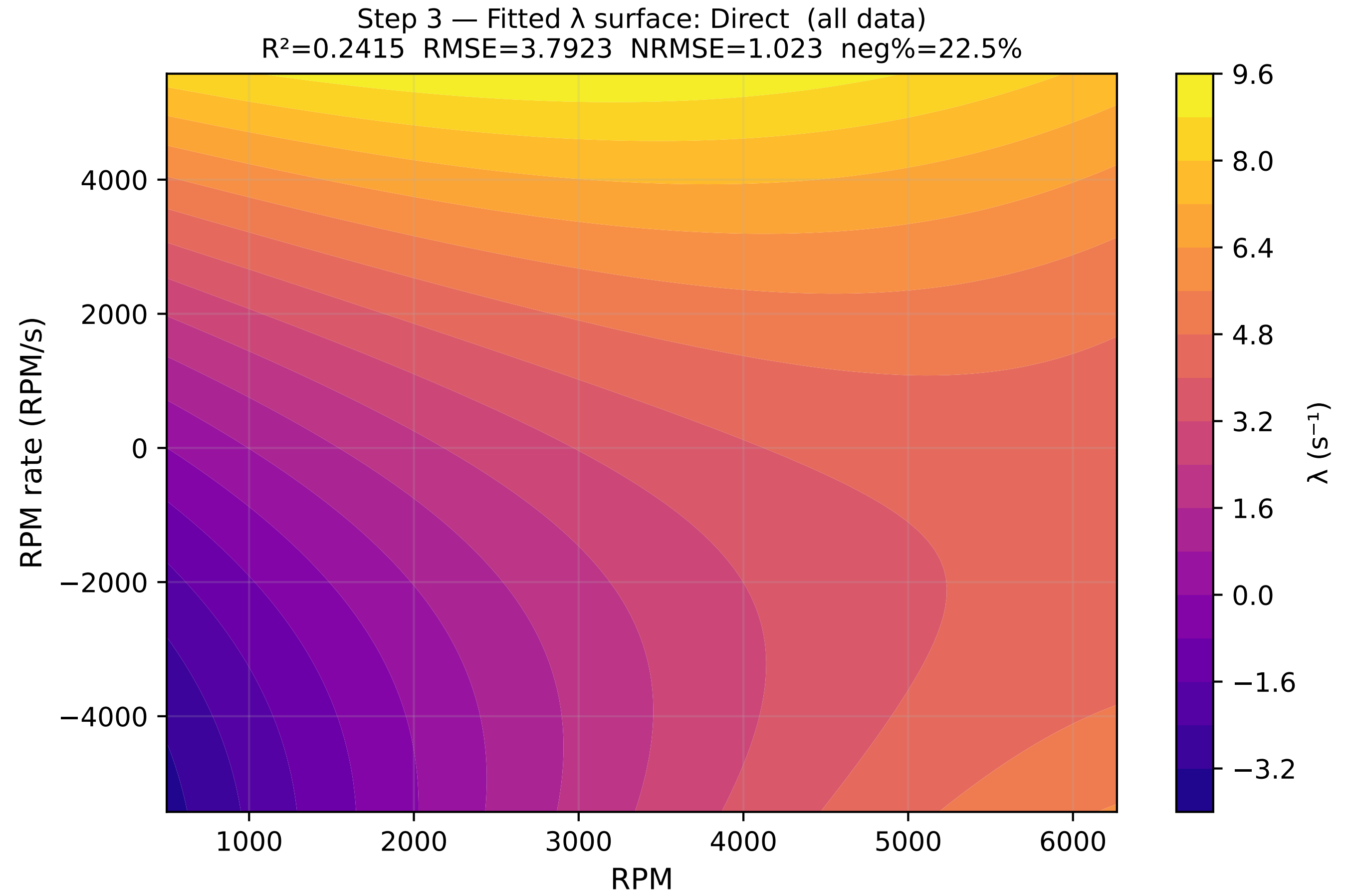

Step 2 — Lag lookup: vs RPM
thrust
torque
Lag (ms)
Mean RPM
180
160
140
120
100
2500
3000
3500
4000
4500
5000
5500
6000
6500
80
60
40
20
0
−20
−40
−60


Step 3 — λ values | RPM_rate = λ·(RPM_cmd − RPM)
Direct (all data)
R²=0.242 NRMSE=1.023 neg%=22.5
Direct (|rate| ≤ 5k)
R²=0.312 NRMSE=1.033 neg%=23.9
Lag (all data)
R²=0.143 NRMSE=0.355 neg%=0.0
Lag (|rate| ≤ 5k)
R²=0.170 NRMSE=0.355 neg%=0.0
RPM
RPM rate (RPM/s)
λ (s⁻¹)

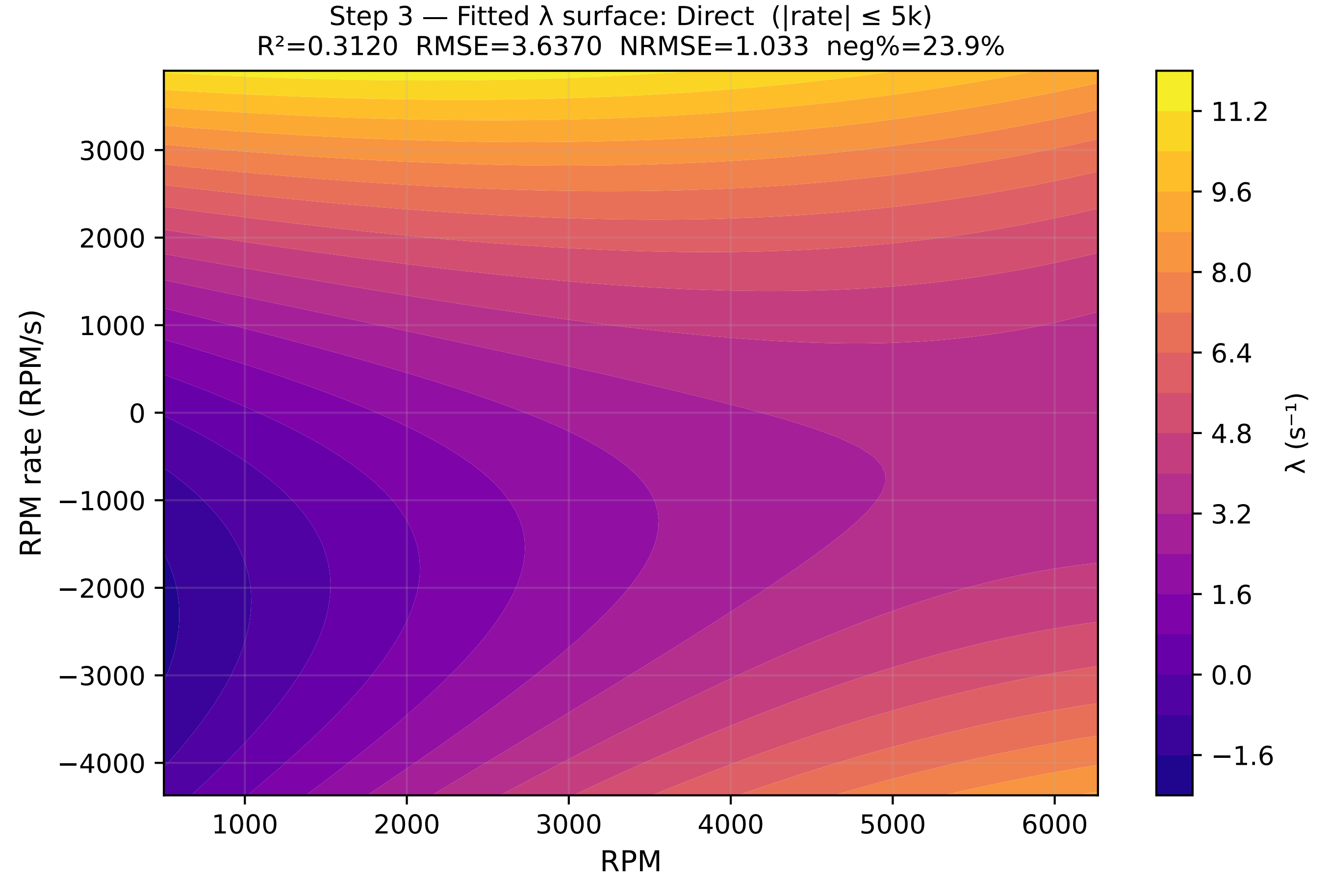


Step 3 — λ residuals (data − fit) | RPM_rate = λ·(RPM_cmd − RPM)
Direct (all data)
R²=0.242 NRMSE=1.023 neg%=22.5
Direct (|rate| ≤ 5k)
R²=0.312 NRMSE=1.033 neg%=23.9
Lag (all data)
R²=0.143 NRMSE=0.355 neg%=0.0
Lag (|rate| ≤ 5k)
R²=0.170 NRMSE=0.355 neg%=0.0
RPM rate (RPM/s)
RPM
residual (s⁻¹)

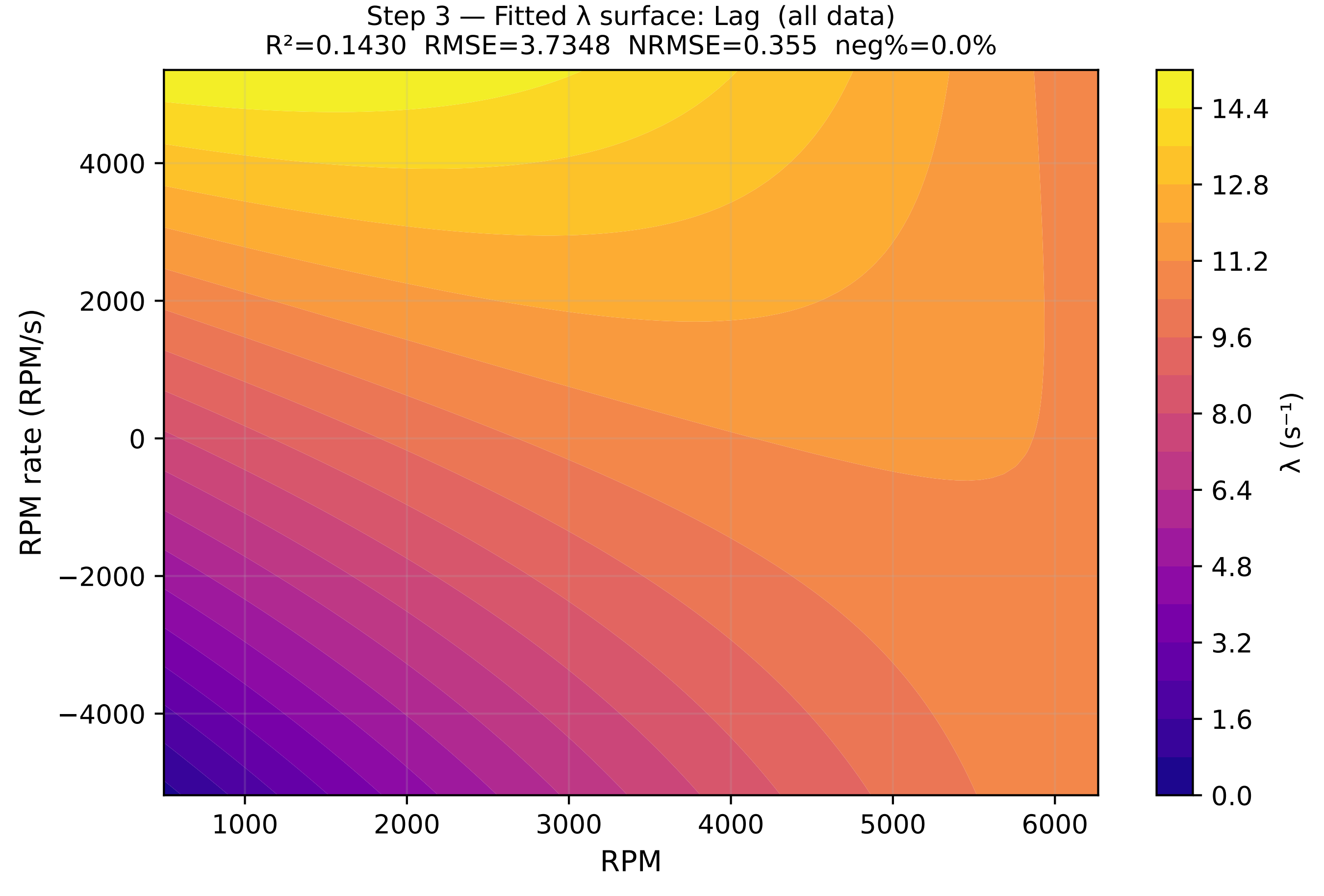

Step 3 — Lambda fit metric comparison
R²
0.242
0.312
0.143
0.170
NRMSE
1.023
1.033
0.355
0.355
RMSE (s⁻¹)
3.792
3.637
3.735
3.728
Neg. fraction
0.225
0.239
0.000
0.000
direct all
direct filtered
lag all
lag filtered



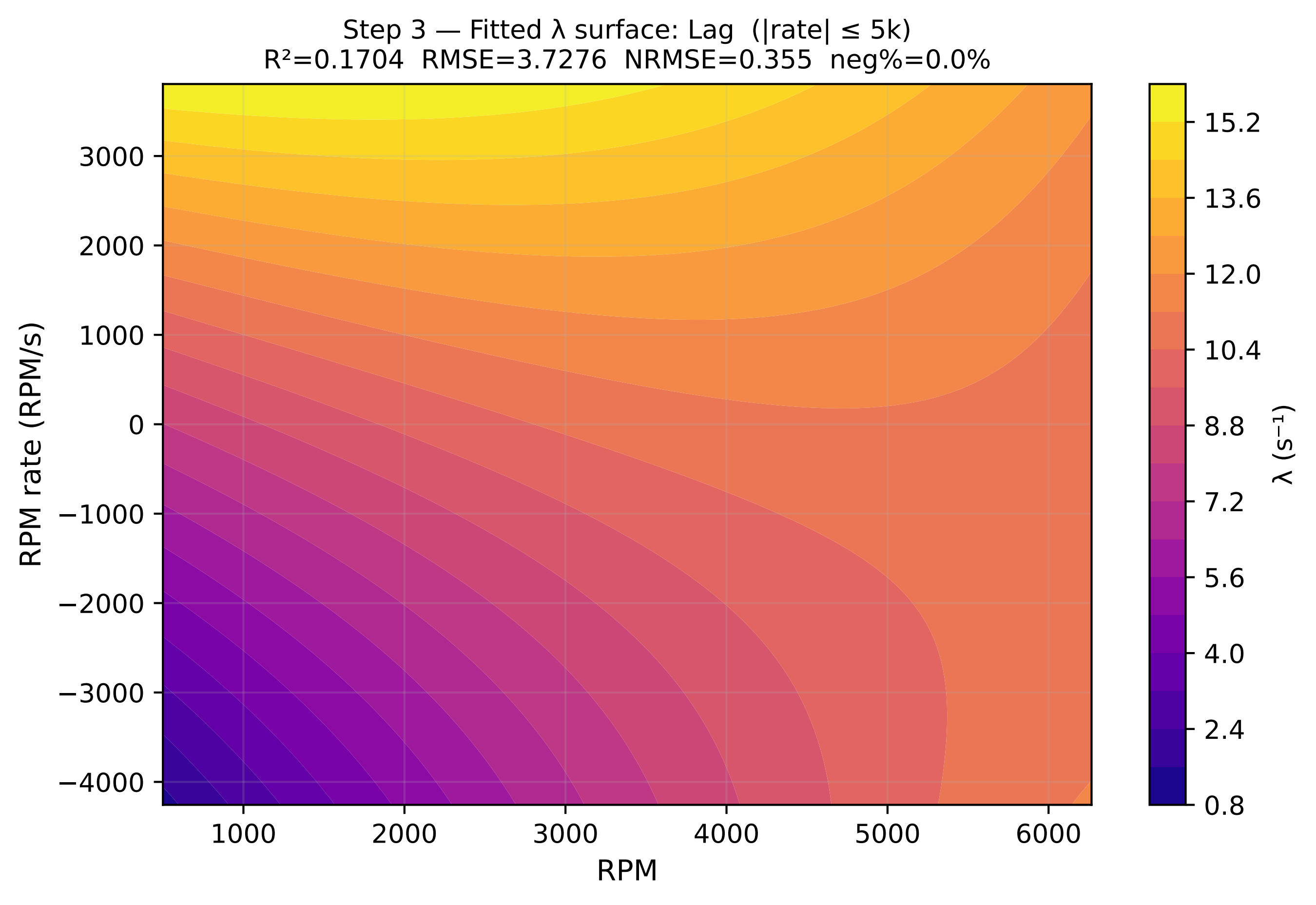
Step 3 — Fitted λ surface: Direct (all data)
R²=0.2415 RMSE=3.7923 NRMSE=1.023 neg%=22.5%
RPM rate (RPM/s)
4000
2000
0
−2000
−4000
1000
2000
3000
4000
5000
6000
RPM
9.6
8.0
6.4
4.8
3.2
1.6
0.0
−1.6
−3.2
λ (s⁻¹)



Step 3 — Fitted λ surface: Direct (|rate| ≤ 5k)
R²=0.3120 RMSE=3.6370 NRMSE=1.033 neg%=23.9%
RPM rate (RPM/s)
3000
2000
1000
0
−1000
−2000
−3000
−4000
1000
2000
3000
4000
5000
6000
RPM
11.2
9.6
8.0
6.4
4.8
3.2
1.6
0.0
−1.6
λ (s⁻¹)



Step 3 — Fitted λ surface: Lag (all data)
R²=0.1430 RMSE=3.7348 NRMSE=0.355 neg%=0.0%
RPM rate (RPM/s)
4000
2000
0
−2000
−4000
1000
2000
3000
4000
5000
6000
RPM
λ (s⁻¹)
14.4
12.8
11.2
9.6
8.0
6.4
4.8
3.2
1.6
0.0



Step 3 — Fitted λ surface: Lag (|rate| ≤ 5k)
R²=0.1704 RMSE=3.7276 NRMSE=0.355 neg%=0.0%
RPM rate (RPM/s)
3000
2000
1000
0
−1000
−2000
−3000
−4000
1000
2000
3000
4000
5000
6000
RPM
15.2
13.6
12.0
10.4
8.8
7.2
5.6
4.0
2.4
0.8
λ (s⁻¹)